Rainfall Comparison between Seattle and Philadelphia

Introduction

This project compares rainfall data from Seattle and Philadelphia to determine which city receives more rainfall annually and experiences a greater proportion of rainy days throughout the year. The null hypotheses are that the mean daily rainfall and the proportion of rainy days are equal between the two cities.

Data

The data analyzed in this project was sourced from the National Oceanic and Atmospheric Administration (NOAA) through the NOAA Climate Data Online (CDO) portal. The data is from two weather stations, one in Seattle and one in Philadelphia. The data spans a total of five years, from January 1, 2018, to December 31, 2022.

The data can also be accessed through the following link:

https://www.ncei.noaa.gov/cdo-web/

Analysis

The data for this project was gathered by NOAA, however, it had to be appropriately cleaned and organized for analysis. The data that we used was the precipitation data gathered by the two weather stations, and the data was stripped of non-relevant data and reorganized into a long format. This data was then visualized and aggregated to find underlying patterns regarding the precipitation data in terms of rainfall per year and the proportion of days with rainfall. On days of the year where rainfall data was missing, the mean rainfall was added in its place.

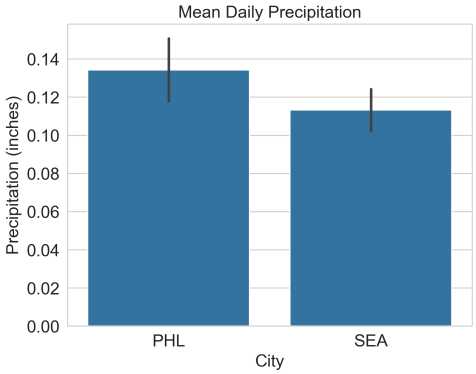
Results

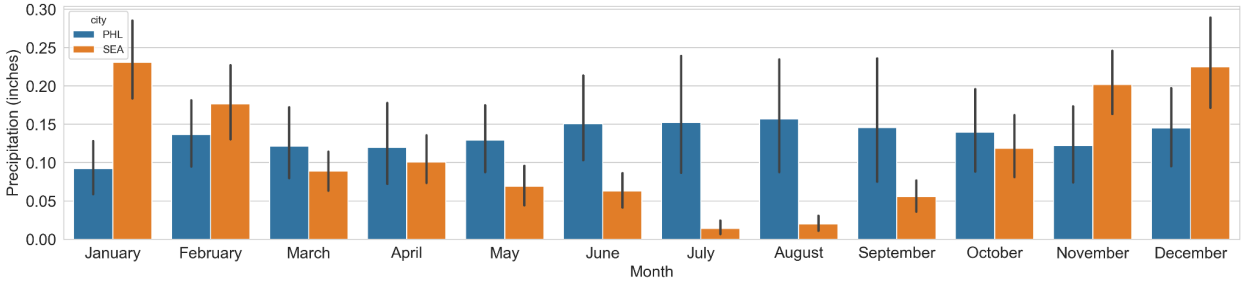
Comparison of Average Rainfall

Our analysis indicated that Philadelphia received more inches of rainfall than Seattle per year, however, Seattle had a higher proportion of rainy days per year when compared to Philadelphia. Philadelphia received, on average, 0.1342 inches of rain per day, while Seattle received an average of 0.1133 inches of rain per day. This reflected that, in terms of inches of rain, Philadelphia received an average of 18.5% more rain per year throughout the 5-year span that our dataset covered. This is reflected in the visuals below.

The visuals also indicate that in the month of January, Seattle averaged 0.2307 inches of rain, while Philadelphia received an average of 0.0923 inches of rain, indicating that Seattle received 150% more inches of rain per day on average in January. In the month of July, Philadelphia averaged 0.1519 inches of rain per day, and Seattle averaged 0.0140 inches of rain per day. This shows that in July, Philadelphia averaged 985% more inches of rain per day when compared to Seattle.

An independent-samples t-test was conducted for each month to determine which months showed a statistically significant difference in mean daily rainfall at the 0.05 significance level. We found that the difference in mean daily rainfall was statistically significant in 8 out of 12 months, meaning that in those months, we could reject the null hypothesis that the two cities received the same amount of rain on average.

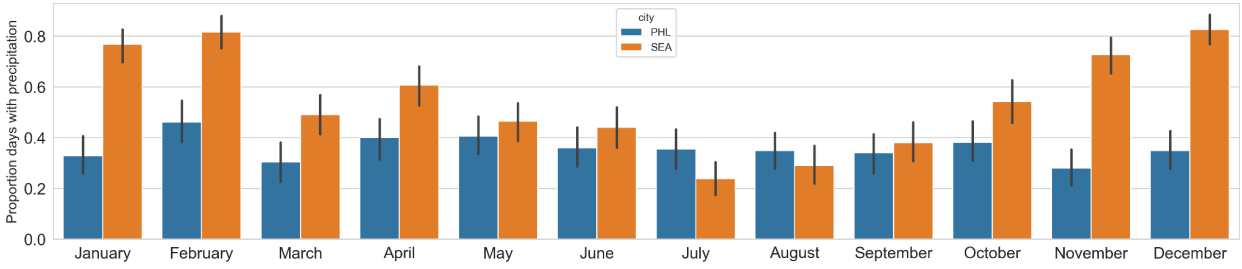


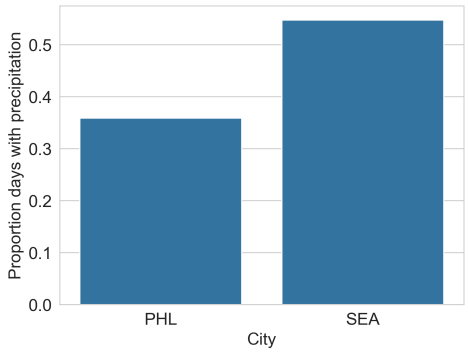


Comparison of Proportion of Rainy Days

When viewed proportionally, on the other hand, Seattle had a higher proportion of rainy days per year when compared to Philadelphia. Throughout the entire year, 54.7% of Seattle's days had rain, while Philadelphia experienced rainy days 35.9% of the time. In January specifically, Seattle experienced rain on 76.8% of days, while Philadelphia experienced rain 32.9% of the time. On the other hand, in the month of July, Philadelphia had rainy days, on average, 35.5% of the time, while Seattle saw rain 23.9% of the time. These results are visualized below.

A two proportion z-test was conducted for each month to determine which months showed a statistically significant difference in their proportion of rainy days at the 0.05 significance level. We found that the difference in mean daily rainfall was statistically significant in 8 out of 12 months, meaning that in those months, we could reject the null hypothesis that the two cities received the same proportion of rainy days.





Conclusion

Seattle and Philadelphia were the focus locations of this project, and the key questions that were answered were which city has a higher mean daily rainfall in terms of inches and which city sees a higher proportion of rainy days throughout the year. We also presented a null hypothesis that both cities have the same daily mean rainfall, and that both cities have the same proportion of rainy days throughout the year.

In terms of mean annual rainfall, Philadelphia averaged 18.5% more rain throughout the year. In terms of the proportion of days with rain, on the other hand, Seattle averaged 52.4% more days with rain throughout the year compared to Philadelphia. These were yearly averages, and there was a lot of variety each month of the year.

When we looked at the monthly patterns in the data, the difference in mean daily rainfall was statistically significant in 8 out of 12 months. The only months that were not statistically significant were February, March, April, and October. This allowed us to reject our null hypothesis that the average daily rainfall in Seattle and Philadelphia was equal for 8 out of 12 months.

When we looked at the monthly patterns in the proportion of rainy days, the difference in the proportions of rainy days was statistically significant in 8 out of 12 months. The months that were not statistically significant were May, June, August, and September. This allowed us to reject our null hypothesis that the proportions of rainy days were equal in Seattle and Philadelphia for 8 out of the 12 months of the year.

This study focused on analyzing precipitation data in Philadelphia and Seattle. However, a limitation of the study is that our data covered only a 5-year period. This may be insufficient to predict future weather patterns, as well as to generalize historical weather patterns. That being said, this study was able to reject our null hypotheses that the mean daily rainfall and the proportion of rainy days were equal in both cities in 8 out of the 12 months.